Does Concept Forgetting Survive Relearni

Measuring whether erased concepts return after ordinary fine tuning

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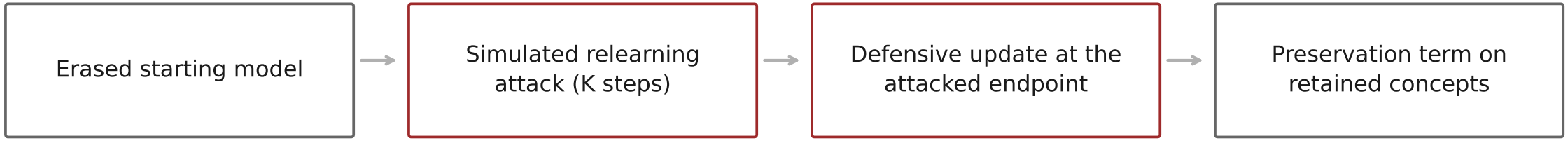
Howard University

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Removing a concept from a released model is useful only if we know how easily later training can bring it back.

Our approach: UTRAP

Anticipate the relearning attack during training, while preserving unrelated capability



repeat for 400 outer updates

K is the number of simulated attack steps.

K = 8 is the configuration under test. K = 0 is the control: no attack is simulated, so the update reduces to erasure plus a preservation term.

Implementation detail that bounds every claim in this talk.

The simulated attack is run with its trajectory detached. We take one gradient of the loss at the attacked endpoint and treat the path to that endpoint as fixed. This is a first order approximation. We do not differentiate through the attack trajectory.

How we evaluate forgetting

Three measurements, one shared protocol

1. Target suppression before attack

Does the model still generate the concept at all?

2. Preservation of unrelated concepts

retain8: closed set accuracy on 8 concepts nobody erased, 120 images. This is not a general image quality measure.

3. Target recovery after relearning: H

Fraction of 15 images, from 3 held out prompts the attacker never trained on, that a fixed 12 way CLIP classifier labels as the target. Denominator 15, so H moves in steps of 0.067. Higher H after an attack means more of the concept returned.

Setting

Base model Stable Diffusion v1.4. Concepts: church (main), Van Gogh style (second).

Attack: fine tuning on generated images of the concept. Budgets 10, 25, 50 and 100 steps. Attacker learning rates 1e-5, 5e-5 and 1e-4.

3 independent defence builds per UTRAP arm, 3 attack seeds per cell.

Validity gate

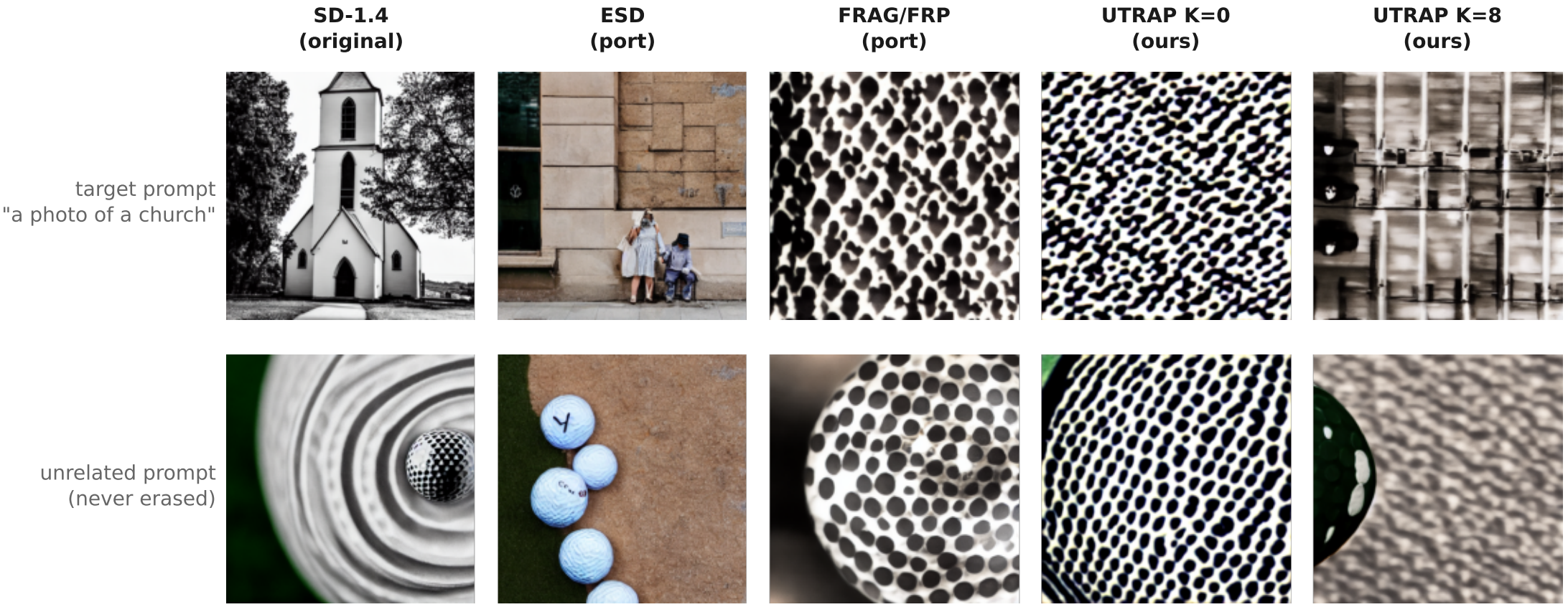
An attack counts only if it demonstrably relearned its own training prompts. Failed gates are reported as inconclusive, never as successful defence.

Provenance labels used throughout

official code execution | from paper port | domain adaptation | our method

What forgetting looks like

Same prompt, same generator seed 0, 25 sampling steps, guidance 7.5, 512x512. Only the weights differ.



Top row: the erased concept. Bottom row: a concept no method targeted, as the specificity control.

The same comparison across every method we hold

Identical generation settings. Method families with known implementation defects are excluded and named below.



Excluded from any claim about a published method:

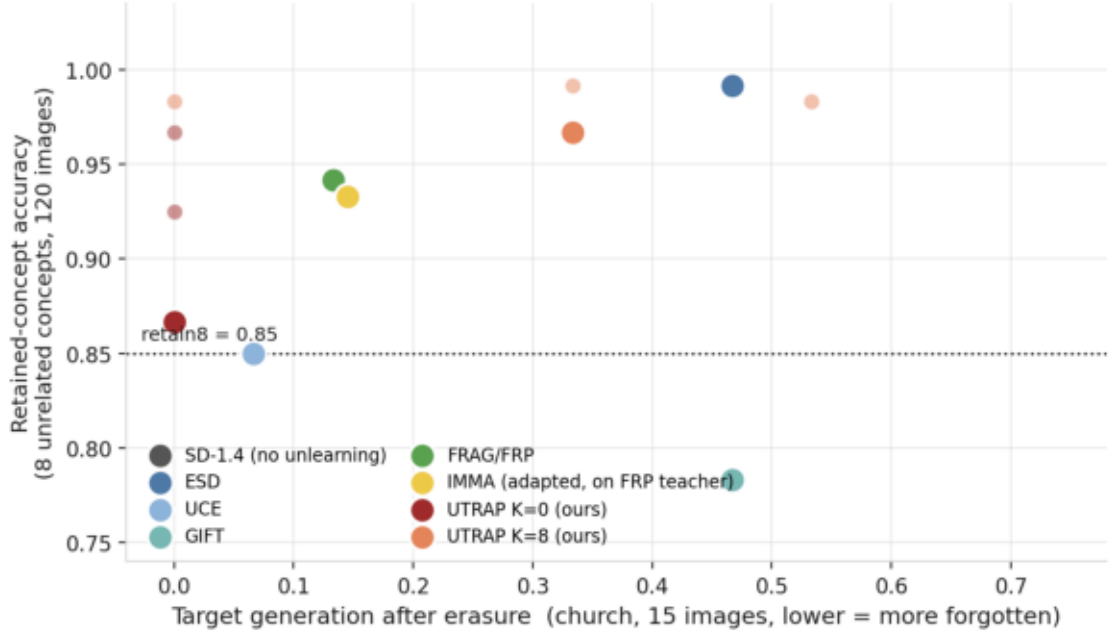
AdvUnlearn: post-attack evaluation loaded the base text encoder, so those numbers are void

Meta-Unlearning: our port omits two of the three terms of the paper's meta objective

WSR: implements the opposite regularisation direction from the paper it cites

Initial forgetting and preservation

Where each method starts, before any attack



Large markers: one representative build per method. Small translucent markers: the additional UTRAP defence builds. Exactly coincident points are nudged apart to stay visible.

Read it as

left = more forgotten

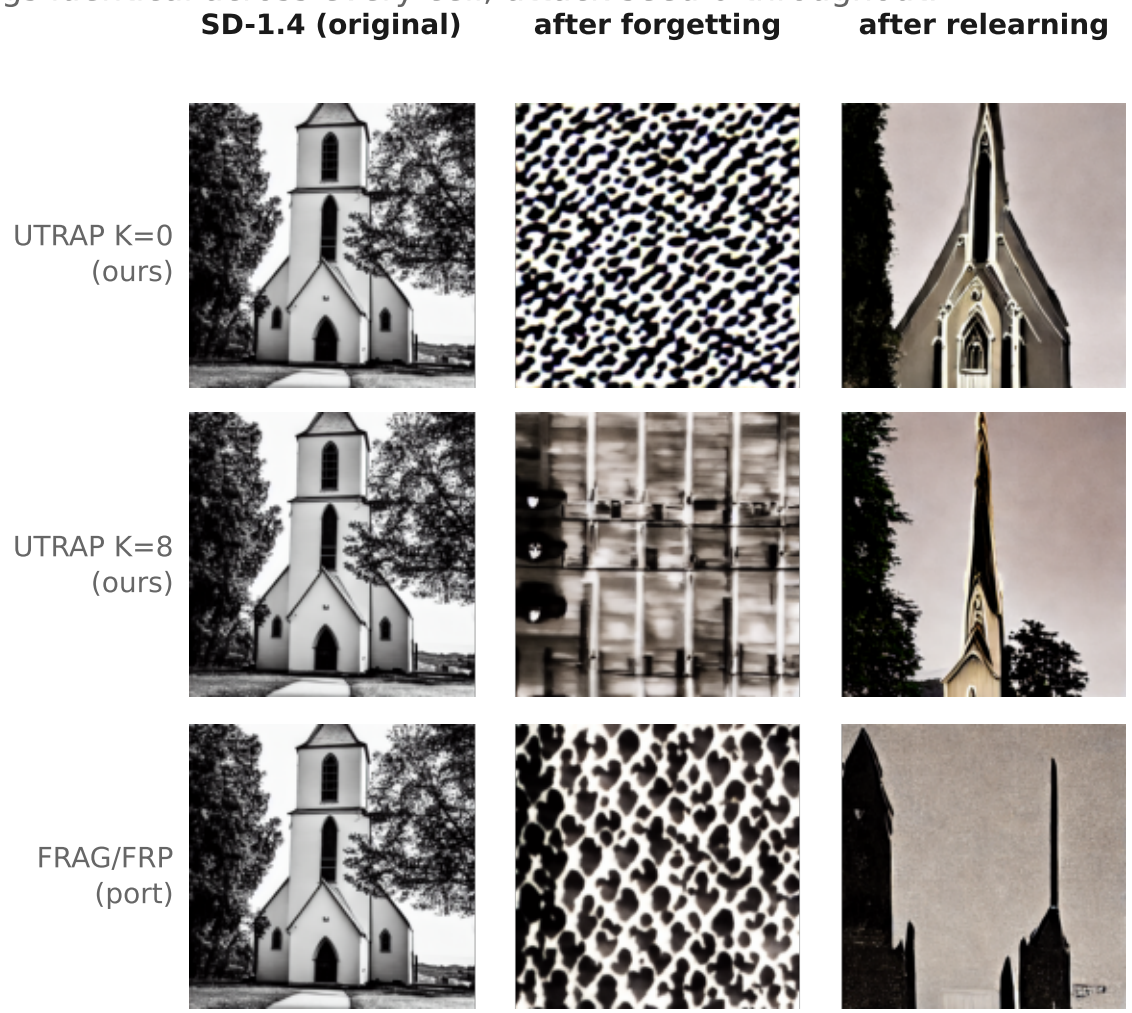
up = more preserved

Why it matters

The methods do not begin the later attack from the same level of forgetting, so post attack numbers cannot be compared without accounting for where each one started.

Does the concept return?

Original model, after forgetting, after a 25 step relearning attack at attacker lr 5e-5. Prompt, generator seed and all generation settings identical across every cell; attack seed 0 throughout.



Recovery H at exactly these settings

25 steps, attacker lr 5e-5, attack seed 0

UTRAP K=0: H = 1.000 (gate valid)

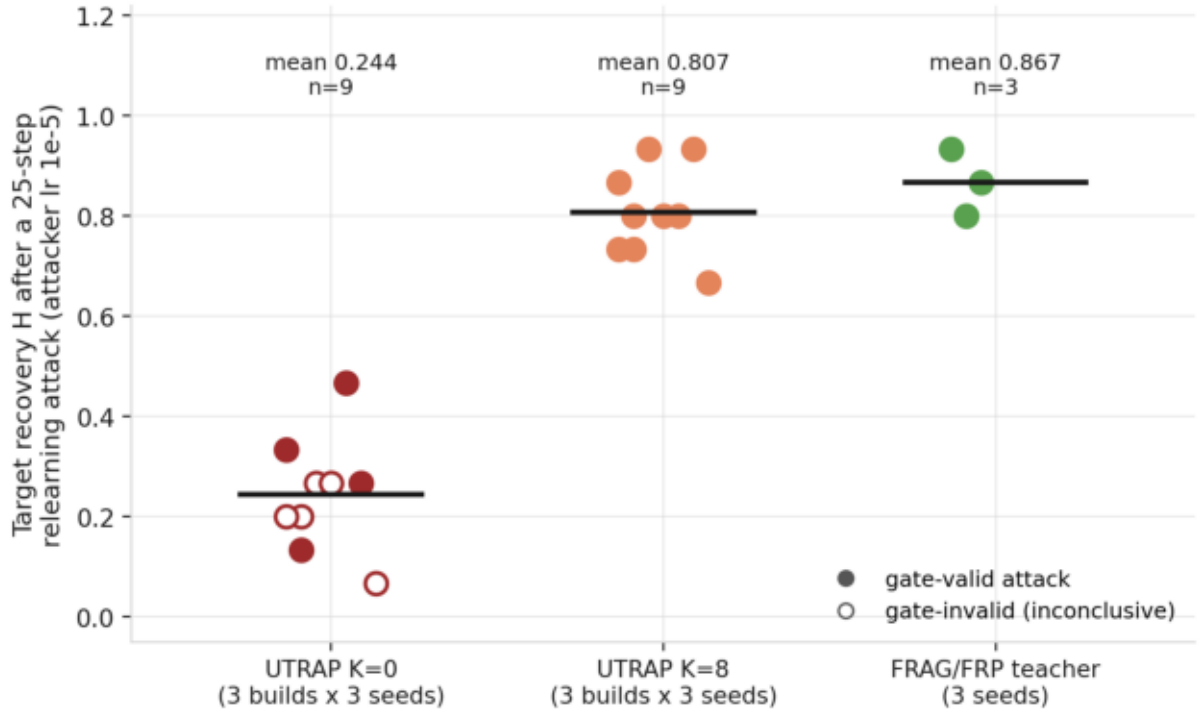
UTRAP K=8: H = 1.000 (gate valid)

FRAG/FRP teacher: H = 1.000 (gate valid)

Across the full sweep every arm reaches H = 1.000 at some tested attacker setting. Worst observed means the maximum over the grid we tested, not a guarantee against a stronger attacker.

Does attack anticipation help?

Fixed outer update comparison. Both UTRAP groups ran 400 outer updates; K=8 additionally received about three times K=0's training samples.



Both UTRAP groups ran 400 outer updates; K=8 additionally received about three times K=0's training samples.

Measured

UTRAP K=8: mean H 0.807 (n=9)

Teacher: mean H 0.867 (n=3)

UTRAP K=0: mean H 0.244 (n=9)

Conclusion

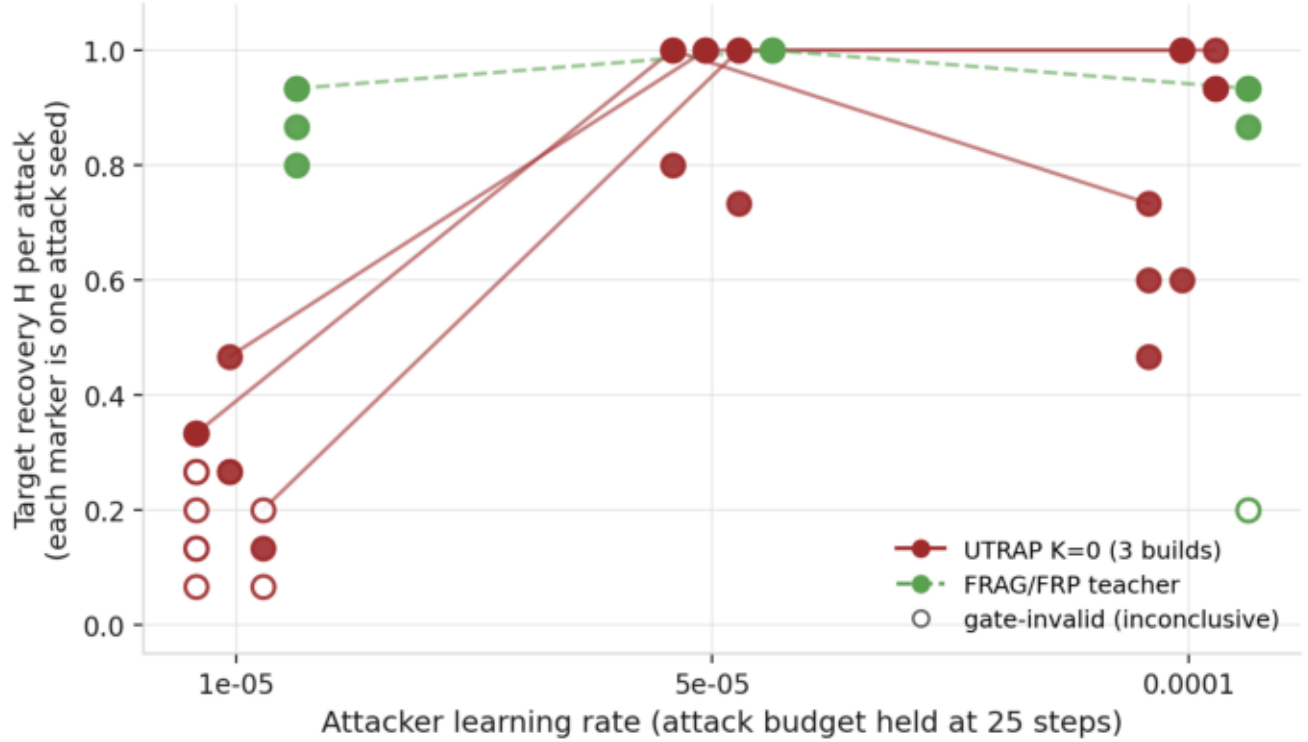
At matched outer updates, and with more data, K=8 is not distinguishable from leaving the teacher untouched. Simulating the attacker gave no measurable benefit here.

Scope

One concept, one teacher, one attack budget, three builds per arm.

Attack calibration changes the conclusion

Same checkpoints, same 25 step budget. Only the attacker learning rate varies.



Lines join the maximum over seeds at each learning rate. Markers are individual attacks; hollow markers failed the validity gate.

The same checkpoint, two attacker settings

UTRAP K=0 build 0, worst observed H

lr 1e-5 -> 0.333

lr 5e-5 -> 1.000

What this means

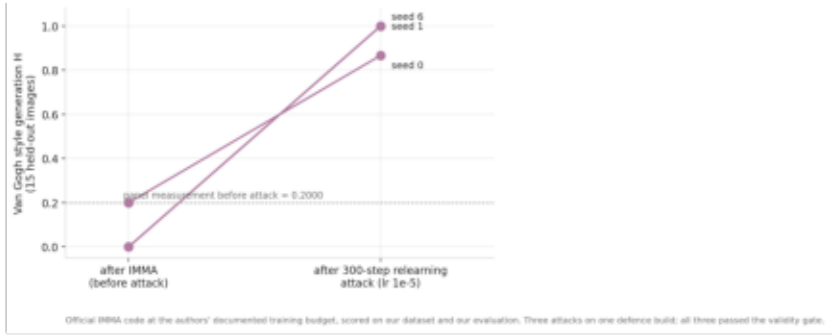
A defence can look resistant because the attacker was badly tuned. A larger learning rate is not automatically a stronger attack: effectiveness peaks and then falls.

Any H measured at a single learning rate is a lower bound on recovery, not a robustness estimate.

A published method under our evaluation: IMMA

Official IMMA code at the authors' documented training budget, on Van Gogh style. Our dataset and our evaluation, not a reproduction of the paper's reported results.

	SD-1.4 (original)	ESD teacher (authors' checkpoint)	after IMMA (official code)
prompt: "a painting in the style of Van Gogh"			



Verified from the records

Before attack: target 0.2000, retain8 0.9083

After a 300 step attack at lr 1e-5, H = 0.867, 1.000, 1.000

3 attacks on ONE defence build. 3 of 3 passed the validity gate.

Official code execution on our benchmark. Not ranked against the church results, which were produced under different conditions.

What we learned, and the next decision

1. Forgetting did not survive relearning, for any method we tested.

Across ESD, FRAG/FRP, UCE, GIFT, adapted IMMA and both UTRAP configurations, a 25 step fine tune restored the concept once the attacker was properly configured.

2. Attack anticipation produced no measurable benefit.

At matched outer updates, with roughly three times the training samples, K=8 was indistinguishable from leaving the teacher untouched, across three independent builds.

3. Apparent robustness tracked attacker tuning, not defence strength.

The same checkpoint moved from apparently resistant to fully recovered when the attacker learning rate changed at an identical step budget. We were misled by this ourselves before sweeping the attacker.

UTRAP does not currently demonstrate a repeatable advantage, and we are not presenting it as one.

Next experiment: tune the attacker against our strongest configuration with the same effort we spend tuning the defence, instead of at a fixed learning rate. If the concept still returns, the conclusion is that this family of defences does not deliver durable forgetting, and that is a result worth reporting on its own.

References

Base model (Stable Diffusion v1.4)

R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer. High-Resolution Image Synthesis with Latent Diffusion Models. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022. DOI: 10.1109/CVPR52688.2022.01042. arXiv:2112.10752

Evaluator (CLIP)

A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever. Learning Transferable Visual Models From Natural Language Supervision. International Conference on Machine Learning (ICML), PMLR 139:8748-8763, 2021. arXiv:2103.00020

ESD

R. Gandikota, J. Materzynska, J. Fiotto-Kaufman, D. Bau. Erasing Concepts from Diffusion Models. IEEE/CVF International Conference on Computer Vision (ICCV), 2023. arXiv:2303.07345. https://erasing.baulab.info/

UCE

R. Gandikota, H. Orgad, Y. Belinkov, J. Materzynska, D. Bau. Unified Concept Editing in Diffusion Models. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024. arXiv:2308.14761. https://unified.baulab.info/

IMMA

A. Y. Zheng, R. A. Yeh. IMMA: Immunizing text-to-image Models against Malicious Adaptation. European Conference on Computer Vision (ECCV), 2024. arXiv:2311.18815

GIFT (preprint)

A. Abdalla, I. Shaheen, D. DeGenaro, R. Mallick, B. Raita, S. A. Bargal. GIFT: Gradient-aware Immunization of diffusion models against malicious Fine-Tuning with safe concepts retention. arXiv preprint arXiv:2507.13598, 18 July 2025. No peer-reviewed venue listed.

Meta-Unlearning

H. Gao, T. Pang, C. Du, T. Hu, Z. Deng, M. Lin. Meta-Unlearning on Diffusion Models: Preventing Relearning Unlearned Concepts. IEEE/CVF International Conference on Computer Vision (ICCV), 2025. arXiv:2410.12777

ResAlign (preprint)

J. Li, W. Gu, Y. Wang, X. Qi, X. Li, Y. Wang, Z. Qin, K. Zhang. Towards Resilient Safety-driven Unlearning for Diffusion Models against Downstream Fine-tuning. arXiv preprint arXiv:2507.16302, 2025. Stated by the authors' repository as accepted to NeurIPS 2025; acceptance not independently verified here.

FRAG / FRP (preprint, implementation unavailable)

Distance Is Not Enough: Forget-Retain Alignment Gap Predicts LLM Relearning Robustness. arXiv preprint arXiv:2608.25429. The public repository styles this as EMNLP 2026 Main Conference; the repository contains a README and two images and ships no implementation (0 Python files), so our FRP arm is a from-paper port with no reference to check against.

MLAC / Self-Destructing Models

P. Henderson, E. Mitchell, C. D. Manning, D. Jurafsky, C. Finn. Self-Destructing Models: Increasing the Costs of Harmful Dual Uses of Foundation Models. AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2023. arXiv:2211.14946

SOPHON

J. Deng, S. Pang, Y. Chen, L. Xia, Y. Bai, H. Weng, W. Xu. SOPHON: Non-Fine-Tunable Learning to Restrain Task Transferability For Pre-trained Models. IEEE Symposium on Security and Privacy (S&P), 2024. arXiv:2404.12699